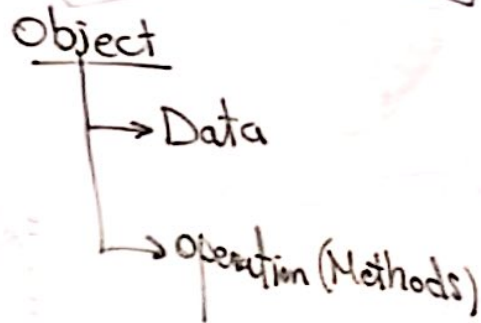
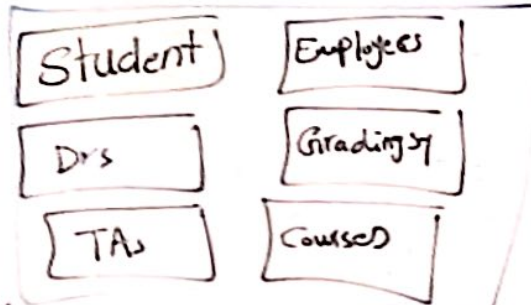


object oriented programming (OOP)

concept of objects

EX: college management system



Student

name
ID
gpa
level
program
change level
change gpa

Product

product name
product code
price
discount
set price
set discount
get product name

Class $\Rightarrow$ template of object

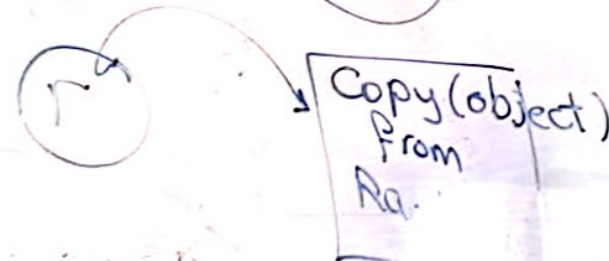
↳ Data
↳ methods

How to take object from the class:

class name. reference variable = new class name ();
memory // نيكو جگہ

EX

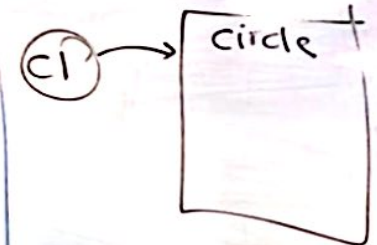
Random r = new Random ();



Scanner input = new Scanner (system.in);



Circle c1 = new Circle ();



Memory

(Y1)

Rectangle

50.0

width

(Y2)

Rectangle

length 40.0

width 0.0

Error
Width
private
cannot
be
accessed

Public class test

```
{ public static void main(String[] args)
{
}
```

```
Rectangle r1 = new Rectangle();
Rectangle r2 = new Rectangle();
```

```
r1.length = 50;
```

```
r2.length = 40;
```

```
r1.width = 15;
```

Public class Rectangle

```
{ public double length;
private double width;
```

Class
Data
Methods

Public class

Data

Methods

UML modeling language

classname
Data
Methods

Data field
attribute

Ex

Rectangle	← class name
length (double)	Data
width (double)	
setLength	Methods


```

public class test
{
    public static void main(String[] args)
    {

```

```

        Rectangle r1 = new Rectangle();
        Rectangle r2 = new Rectangle();
        r1.setlength(50);
        r2.setlength(5);
        r1.setwidth(4);

```

Sy — (r1.getlength());

Sy — (r1.getwidth());

```

public class Rectangle
{
    private double length;
    private double width;

```

```

    public void setlength(double L)
    {
        if (L > 10)
            Length = L;
        else
            sy — ("invalid value");
    }

```

```

    public void setwidth(double w)
    {
        width = w;
    }

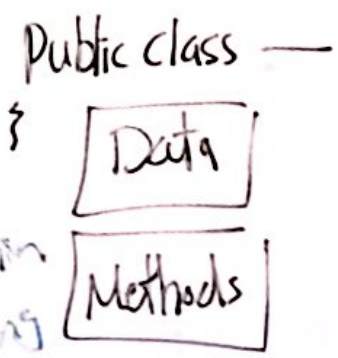
```

```

    public double getlength()
    {
        return length;
    }
    public double getwidth()
    {
        return width;
    }

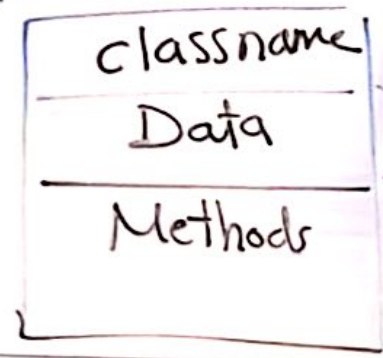
```

Class
Data
Methods



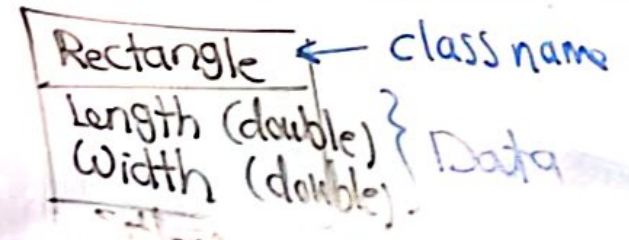
Linear relation
Data hiding

UML (Unified modeling language)



Data Hiding

Ex



← class name

Data